Comments to the Author

Thank you for a thorough revision. You acted on the previous comments rather than arguing with them: the test-selected policy is gone, ten seeds replace three, the multi-fold analysis is now primary, the baselines and constant control are in, and partition 10 is correctly reported as a null. I also checked the numbers, every fold mean in Table S1 reproduces Table I, and the abstract figures follow exactly. That is not common and made the paper easy to review.

A few things still need attention. None of them require new measurements.

1. Are the six partitions the six measurement days? The main text says the partitions come from Combination identifiers and are not acquisition days. The supplement says the protocol ran on six separate days. Since 378 sequences divide evenly into both, it is unclear whether they are the same thing. If they are not, every split shares sessions between training and test, and "held-partition validation" claims more than it delivers. You appear to have the day labels could you rerun the primary protocol as leave-one-day-out? It is a re-split of data you already have, and if the gains survive it, your main claim becomes much stronger. If the identifiers do not allow it, please say so directly.

2. At full budget your model is not the best on any target. Section V-E gives two examples, but reading Table IV across all four: generic SSL beats you on NH_3 (16.13 vs 17.36), SVR on H_2 (31.37 vs 31.60), Extra Trees on ethylene (1.398 vs 1.627), and PLSR on CO by 1.86× (25.57 vs 47.45). Three of these are ties within one SD; CO is not. Please state this plainly in the abstract and conclusion. The CO gap also deserves a sentence of explanation, since a linear model beating a Transformer by almost 2 times on your best-gain target is worth understanding.

3. Please measure label efficiency in labels. Everything is currently reported as percentage MAE reduction at a fixed budget. Table II already contains what a user needs: fine-tuned at 10 pairs roughly equals scratch at 20 for NH_3, CO and ethylene, and less than that for H_2. A small table or one figure with both curves would make the claim concrete.

4. Humidity and temperature are still missing. For SnO_2 sensors, RH shifts both baseline and sensitivity, and across six measurement days it is exactly the confound your evaluation is meant to test. Was RH controlled, recorded, or neither? Any of those answers is fine; the silence is not.

5. NH_3 shows no benefit at five pairs (86.39 → 86.02, against an SD of 36 ppm), while the other three gases do gain there. Since the small-budget case is where label efficiency matters most, please comment on why. It would also help to say that an 86 ppm error on a 0–500 ppm range is not usable quantitatively, whatever the policy.

6. The H_2 gain depends on the window. Table VI gives 21.5% at the early window, a reversal at the middle window (33.99 → 35.60), and 6.8% at full exposure. The spread across window placements is larger than the transfer effect. Predefining the early window was correct, but please connect Table VI to the abstract number so readers know the headline figure is conditional on that choice.

7. Table II uses the farthest-point design, but does not say so. This matters because Table S3 shows random is significantly *better* at 40 pairs for both NH_3 (−4.46 ppm) and H2 (−8.32 ppm). Please state the design in the caption and describe the design effect as budget-dependent as well as target-dependent.

8. The four primary intervals are unadjusted, and NH_3 is marginal at [0.38, 8.30]. Under any correction for four comparisons it would likely include zero. NH_3 is also the gas where sensor ID and attention show no benefit. One sentence acknowledging NH_3 as the weakest case throughout would be consistent with the rest of the paper. Related: the CO/ethylene bootstrap resamples only three folds please mark those intervals as indicative.

9. Transfer reverses in fold 4 (NH_3 16.22 → 18.83; H2 25.70 → 26.36), which a reader going to the supplement will find. A brief acknowledgment, and any explanation, would help.

10. Your sampling rate is better supported than you say. You disclaim an unverified sampling frequency, but 162,404 observations over 378 sequences gives 429.6 samples per sequence, and 880 s ÷ 429.6 = 2.048 s matching your stated 2.0 s to within 2.5%. You can drop the caveat and describe the window physically: 64 samples ≈ 131 s, about the first 73% of the 180 s exposure. That is far more useful to a sensor reader than a sample index.

Smaller points

- Z is used both for the sensor-token matrix in Eq. (8) and the mask indicator in Eqs. (11) and (13). Please rename one. The span start a_k and the batch index b are also undefined.
- Eq. (14) guards the first denominator but not $\sum M$ in the second.
- Page 4 has a literal placeholder: Supplementary NH_3/H_2 dataset and acquisition details?
- The third-person "the authors" reads awkwardly throughout, including the abstract. "We," or consistent passive voice, would be better. Also "avoids need for" → "avoids the need for."
- The abstract is ~290 words and mostly caveats. Lead with the result and protocol, keep one or two sentences of scope, and leave the rest to the discussion, the correction from last round has overshot.
- Table V's CO ablation entries (10.27 ± 12.18, 7.93 ± 14.86) are consistent with no effect; please say so for CO specifically.
- The repository link sits at the end of the biographies; it belongs in a data and code availability statement.
- Please state the pretraining compute cost once, so readers can weigh labels saved against compute spent.